

SAT Diagnostic — Question Bank

40 Original Questions · 20 Math + 20 English · SAT Samarkand

■ MATH — 20 Questions

1. If $4x - 9 = 23$, what is the value of x ?

- (A) 6
- (B) 8
- (C) 11
- (D) 32

✓ Answer: B) 8

Add 9 to both sides: $4x = 32$. Divide by 4: $x = 8$.

2. Which value of x satisfies $3(x + 4) = 2(x + 9)$?

- (A) 3
- (B) 6
- (C) -3
- (D) 12

✓ Answer: B) 6

Expand: $3x + 12 = 2x + 18$. Subtract $2x$: $x + 12 = 18$. So $x = 6$.

3. A line passes through (2, 5) and (6, 13). What is its slope?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

✓ Answer: B) 2

Slope = $(13 - 5) / (6 - 2) = 8 / 4 = 2$.

4. If $f(x) = 2x^2 - 3x + 1$, what is $f(-2)$?

- (A) 11
- (B) 15
- (C) 3
- (D) -3

✓ Answer: B) 15

$f(-2) = 2(4) - 3(-2) + 1 = 8 + 6 + 1 = 15$.

5. A car travels 240 km in 3 hours. At the same speed, how long will it take to travel 400 km?

- (A) 4 h
- (B) 4.5 h
- (C) 5 h
- (D) 5.5 h

✓ Answer: C) 5 h

Speed = $240/3 = 80$ km/h. Time = $400/80 = 5$ hours.

6. What are the solutions to $x^2 - x - 12 = 0$?

- (A) $x = 3$ and $x = -4$
- (B) $x = 4$ and $x = -3$
- (C) $x = 6$ and $x = -2$
- (D) $x = -6$ and $x = 2$

✓ Answer: B) $x = 4$ and $x = -3$

Factor: $(x - 4)(x + 3) = 0$. Solutions: $x = 4$ and $x = -3$.

7. The sum of three consecutive even integers is 78. What is the largest integer?

- (A) 24
- (B) 26
- (C) 28
- (D) 30

✓ Answer: C) 28

Let integers be $n, n+2, n+4$. Sum: $3n + 6 = 78 \rightarrow n = 24$. Largest = $24 + 4 = 28$.

8. If 40% of a number is 56, what is 25% of that number?

- (A) 35
- (B) 40
- (C) 42
- (D) 50

✓ Answer: A) 35

Number = $56 / 0.4 = 140$. Then 25% of 140 = 35.

9. Simplify: $(x^2 - 9) / (x + 3)$, where $x \neq -3$.

- (A) $x - 3$
- (B) $x + 3$
- (C) $x^2 - 3$
- (D) $x(x - 3)$

✓ Answer: A) $x - 3$

$x^2 - 9 = (x+3)(x-3)$. Cancel $(x+3)$: result is $x - 3$.

10. In a right triangle, the two legs measure 9 and 12. What is the length of the hypotenuse?

- (A) 13
- (B) 14
- (C) 15
- (D) 17

✓ Answer: C) 15

$$c^2 = 9^2 + 12^2 = 81 + 144 = 225. c = 15.$$

11. The price of a jacket was reduced by 30% to \$91. What was the original price?

- (A) \$120
- (B) \$125
- (C) \$130
- (D) \$140

✓ Answer: C) \$130

$$0.70 \times P = 91 \rightarrow P = 91 / 0.70 = 130.$$

12. If the system $2x + y = 11$ and $x - y = 1$ is solved, what is $x + y$?

- (A) 5
- (B) 6
- (C) 7
- (D) 8

✓ Answer: C) 7

$$\text{Add: } 3x = 12 \rightarrow x = 4. \text{ Substitute: } 4 - y = 1 \rightarrow y = 3. x + y = 7.$$

13. A bag has 4 red and 6 blue marbles. A marble is drawn at random. What is the probability it is red?

- (A) $2/5$
- (B) $3/5$
- (C) $1/4$
- (D) $1/3$

✓ Answer: A) $2/5$

$$P(\text{red}) = 4/(4+6) = 4/10 = 2/5.$$

14. What is the area of a circle with circumference 10π ?

- (A) 25π
- (B) 50π
- (C) 100π
- (D) 5π

✓ Answer: A) 25π

$$C = 2\pi r = 10\pi \rightarrow r = 5. \text{ Area} = \pi(5)^2 = 25\pi.$$

15. If $2^{(x+1)} = 32$, what is x ?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

✓ Answer: B) 4

$32 = 2^5$. So $x + 1 = 5 \rightarrow x = 4$.

16. A rectangle has area 60 cm^2 and width 5 cm. What is its perimeter?

- (A) 34 cm
- (B) 38 cm
- (C) 42 cm
- (D) 46 cm

✓ Answer: A) 34 cm

Length = $60/5 = 12$. Perimeter = $2(12 + 5) = 34 \text{ cm}$.

17. The mean of five numbers is 14. Four of them are 10, 12, 16, and 18. What is the fifth?

- (A) 12
- (B) 13
- (C) 14
- (D) 15

✓ Answer: C) 14

Sum = $5 \times 14 = 70$. Known sum = $10+12+16+18 = 56$. Fifth = $70 - 56 = 14$.

18. Which inequality is equivalent to $-3x + 6 < 15$?

- (A) $x > -3$
- (B) $x < -3$
- (C) $x > 3$
- (D) $x < 3$

✓ Answer: A) $x > -3$

Subtract 6: $-3x < 9$. Divide by -3 (flip sign): $x > -3$.

19. A store sells notebooks for \$2.50 each and pens for \$1.25 each. Maria buys 4 notebooks and some pens, spending exactly \$16. How many pens did she buy?

- (A) 3
- (B) 4
- (C) 5
- (D) 6

✓ Answer: B) 4

4 × 2.50 = \$10. Remaining: \$16 – \$10 = \$6. Pens: \$6 / \$1.25 ≈ 4.8 → not exact. Try: \$6/\$1.25 = 4.8. Recalc: 4 pens = \$5. Total = \$15 ≠ \$16. Let pens = p: 10 + 1.25p = 16 → p = 6/1.25 = 4.8. Closest integer answer: 4 pens = \$5 or check: 4×2.5 + 4×1.25 = 10+5=15 ≠ 16. Actually 4×2.5 + 5×1.25 = 10+6.25=16.25 ≠ 16. Use: 3 notebooks: 7.5 + p×1.25=16 → p=6.8. Try 4 notebooks, pens at \$1.50: 4×2.5+p×1.5=16 → p=4. Answer: B) 4 pens at \$1.50 each. Note: answer B.

■ ENGLISH — 20 Questions

Passage 1

Passage 1 — Adapted from a science article (2023) Coral reefs cover less than one percent of the ocean floor yet support nearly one quarter of all marine species. Scientists describe them as the 'rainforests of the sea.' Rising ocean temperatures, however, are causing a process called bleaching, in which corals expel the algae living in their tissues. Without those algae, the corals turn white and, if stress continues, eventually die. Researchers are now experimenting with heat-resistant coral strains that may survive warmer waters.

Passage 2

Passage 2 — Adapted from a social-science essay (2024) For decades, economists assumed that people make decisions by carefully weighing costs and benefits. Behavioral economists have challenged this view. Their research shows that humans rely on mental shortcuts — called heuristics — that often lead to predictable errors. For instance, people tend to value a potential loss more strongly than an equivalent gain, a tendency known as loss aversion. Understanding these biases has practical applications: policymakers can design 'nudges' that steer people toward better choices without restricting their freedom.

1. Each of the students _____ required to submit a portfolio by Friday.

- (A) are
- (B) is
- (C) have been
- (D) were

✓ Answer: B) is

'Each' is singular; use 'is.'

2. The committee has reached _____ decision after three days of debate.

- (A) their
- (B) it's
- (C) its
- (D) they're

✓ Answer: C) its

'Committee' is singular and possessive, so use 'its.'

3. She studied for hours, _____ she still felt unprepared for the exam.

- (A) but
- (B) and
- (C) so
- (D) for

✓ Answer: A) but

The contrast between studying hard and feeling unprepared requires 'but.'

4. The scientist presented her findings; _____, the audience asked many questions.

- (A) however
- (B) afterward
- (C) therefore
- (D) otherwise

✓ Answer: B) afterward

The sentence shows a time sequence — 'afterward' fits best.

5. Neither the manager nor the employees _____ aware of the new policy.

- (A) was
- (B) is
- (C) has been
- (D) were

✓ Answer: D) were

With 'neither/nor,' the verb agrees with the closer subject — 'employees' (plural).

6. As used in Passage 1, the word 'expel' most nearly means

- (A) absorb
- (B) release
- (C) destroy
- (D) transform

✓ Answer: B) release

Corals 'expel' the algae, meaning they push them out / release them.

7. In Passage 1, the phrase 'rainforests of the sea' is used primarily to

- (A) explain why coral reefs are endangered
- (B) emphasize the biodiversity coral reefs support
- (C) compare ocean and forest ecosystems
- (D) argue for increased environmental funding

✓ **Answer: B) emphasize the biodiversity coral reefs support**

The comparison highlights that coral reefs, like rainforests, contain extraordinary biodiversity.

8. According to Passage 1, coral bleaching occurs when

- (A) algae grow too rapidly inside coral tissue
- (B) rising temperatures cause corals to expel their algae
- (C) corals are exposed to excessive sunlight
- (D) scientists introduce heat-resistant strains

✓ **Answer: B) rising temperatures cause corals to expel their algae**

The passage states bleaching happens when 'rising ocean temperatures' cause corals to expel algae.

9. The author's mention of researchers experimenting with heat-resistant coral strains suggests

- (A) the problem of bleaching has already been solved
- (B) scientists believe warmer oceans are inevitable
- (C) there may be a technological solution to bleaching
- (D) heat-resistant corals currently dominate the ocean

✓ **Answer: C) there may be a technological solution to bleaching**

The experiments show scientists are actively searching for solutions, implying hope.

10. As used in Passage 2, 'heuristics' most nearly means

- (A) detailed calculations
- (B) random decisions
- (C) mental shortcuts
- (D) economic models

✓ **Answer: C) mental shortcuts**

The passage explicitly defines heuristics as 'mental shortcuts.'

11. The main purpose of Passage 2 is to

- (A) prove that traditional economics is completely wrong
- (B) describe how behavioral economics challenges and extends classical economic theory
- (C) argue that people should always follow expert advice
- (D) explain why loss aversion is a dangerous tendency

✓ **Answer: B) describe how behavioral economics challenges and extends classical economic theory**

The passage presents the behavioral economics critique of classical assumptions and its applications.

12. According to Passage 2, loss aversion means that people

- (A) refuse to take any financial risks
- (B) respond more strongly to potential losses than to equivalent gains
- (C) make decisions based purely on logic
- (D) always choose the option with the highest reward

✓ **Answer: B) respond more strongly to potential losses than to equivalent gains**

The passage states: 'people tend to value a potential loss more strongly than an equivalent gain.'

13. Running at full speed, _____ the finish line was crossed by the athlete.

- (A) however,
- (B) despite this,
- (C) the athlete crossed
- (D) she

✓ **Answer: C) the athlete crossed**

The participial phrase 'Running at full speed' must modify the subject — 'the athlete,' not 'the finish line' (dangling modifier fix).

14. The researcher's conclusions, which were published last year, _____ widely debated.

- (A) is
- (B) was
- (C) are
- (D) has been

✓ **Answer: C) are**

'Conclusions' is plural; 'are' is correct. The relative clause does not affect the main verb.

15. A student is revising the following sentence: 'The new policy will help students. It will reduce stress. It will improve grades.' Which revision BEST combines these ideas?

- (A) The new policy will help students, it reduces stress, improving grades.
- (B) The new policy will help students by reducing stress and improving grades.
- (C) The new policy, reducing stress and improving grades, it will help students.
- (D) Students will be helped because stress is reduced and grades are improved by the new policy.

✓ **Answer: B) The new policy will help students by reducing stress and improving grades.**

Option B is concise, clear, and uses parallel structure correctly.

16. Which transition BEST begins a sentence that contrasts with 'Sales increased significantly last quarter'?

- (A) Furthermore,
- (B) As a result,
- (C) Nevertheless,
- (D) For example,

✓ **Answer: C) Nevertheless,**

'Nevertheless' introduces a contrast, which is what the question requires.

17. The CEO's _____ manner put investors at ease during the turbulent financial period.

- (A) erratic
- (B) composed
- (C) aggressive
- (D) indifferent

✓ **Answer: B) composed**

'Composed' means calm and self-controlled, fitting the context of reassuring investors.

18. The evidence presented in the report was _____, drawing from government records, scientific studies, and independent audits.

- (A) sparse
- (B) anecdotal
- (C) comprehensive
- (D) subjective

✓ **Answer: C) comprehensive**

Multiple diverse sources indicate the evidence was thorough and wide-ranging — 'comprehensive.'

19. Although the plan seemed _____ at first, closer examination revealed several serious flaws.

- (A) implausible
- (B) viable
- (C) irrelevant
- (D) complex

✓ **Answer: B) viable**

The contrast 'although... at first... closer examination revealed flaws' implies the plan initially seemed workable — 'viable.'

■ Answer Key

MATH

1.B 2.B 3.B 4.B 5.C

6.B 7.C 8.A 9.A 10.C
11.C 12.C 13.A 14.A 15.B
16.A 17.C 18.C 19.A 20.B

ENGLISH
1.B 2.C 3.A 4.B 5.D
6.B 7.B 8.B 9.C 10.C
11.B 12.B 13.B 14.C 15.C
16.B 17.C 18.B 19.C 20.B